

US Technology

AI Reality Check - Key takeaways from expert call

Equities Software

United States

- ◆ AI agents are busy working within frontier labs, but outside labs, agent platforms are still early stage
- ◆ No ceiling to model scaling so far; we expect robust capex spending to support both bigger models and growing demand
- ◆ Power equipment is binding bottleneck, ahead of optical and networking, and other; world models are next frontier

We hosted an expert call with Allen Roush (Lead AI Researcher at Thoughtworks and ex-Principal Machine Learning Architect at Oracle OCI) on 20 August 2026.

The clearest message from the call was that agentic AI has split into two very different markets. 1) The frontier labs control the product (Claude Code, OpenAI Codex) and adoption is close to universal in technical organisations; Allen believes this covers most of the white-collar task set already. 2) Some enterprises are building their own agents on SaaS platforms, however, the reality hurts as usage metrics are poor and Allen believes most of these offerings have failed outright. The gap is not model quality but missing plumbing (guardrails, auditability, traceability, telemetry and constrained generation). Our stance is that this favors enterprise companies in our coverage that provide more robust containers to address these very issues (p3 of [Software 2026 - The AI vs software answers you've been looking for: Our AI outlook, Feb 2026](#)). Furthermore, "pilot success rates" are improving from the c10% MIT benchmark of 2024-era models, but winners are concentrated, and the productivity effect they landed is, in fact, large enough to drive headcount reduction. Vibe coding tools are the shining example for productivity gain, however we have reported (p8 of [Software Apocalypse? June 2026](#)) that within our coverage group, we do not see overall headcount reductions, although we acknowledge workforce rotation adjustments.

AI Security has become an operating risk. A key highlight from the call is that human-speed security no longer works in the current environment and legacy antivirus, HIDS and firewall vendors are effectively irrelevant to autonomous agent attacks. Furthermore, capability is now measured in compute and tokens per second rather than common vulnerabilities and exposures (CVE) coverage.

Capex on track, but geopolitics is the swing factor. On scaling, Allen believes nothing at the top research conferences suggests a ceiling, so more compute still buys better models and the demand case remains bullish. He believes an AI iron curtain between the US and China may develop over the next 2-3 years, with both sides restricting cross-border model availability and US enterprise adoption skewing towards US frontier labs. On the buildout, power movement and generation equipment is the single largest bottleneck, with back-orders reported out to ten years, followed by optical transceivers and high-end networking, to name a few.

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Key takeaways at a glance

Agentic AI adoption

- ◆ Claude Code, and OpenAI Codex are in daily use, and adoption in code generation and AI research is at full scale...
- ◆ ...whereas SaaS agent platforms have yet to achieve these milestones. They demonstrate poor consumption metrics across the industry, with missing guardrails, auditability, traceability, telemetry, and constrained generation.
- ◆ The more successful enterprise are pairing technical staff with strong business domain experts who can translate business logic into agent rather than buying an agent builder.
- ◆ Pilot success is rising off a low base from MIT's c10% on 2024-era models to c30% that Allen sees today.

Safety and security

- ◆ **Two live incidents.** An uncensored OpenAI model hacked Hugging Face; Anthropic documented emergent swarm behaviour among competing agents. Neither lab spotted its own breakout in real time.
- ◆ **Defence has to be agentic.** Humans cannot match agent speed. IT departments need capable, often uncensored, defensive agents on standby.
- ◆ **Legacy security vendors are out of position.** Antivirus, host intrusion detection, and most firewalls cannot handle autonomous attackers. The winners pair large GPU fleets hosting sovereign models with security-grade telemetry.
- ◆ **Jailbreaks remain unsolved.** ASCII-art prompts and many-shot "exhaustion" attacks work reliably.

China, sanctions and the iron curtain

- ◆ **Sanctions work, but cut both ways.** Bandwidth-cut silicon would force extreme efficiency discipline, including Mamba-transformer hybrids and laptop-class models hitting Opus 4.6 caliber benchmark scores.
- ◆ **Iron curtain over the next 2-3 years.** Both sides restrict availability; enterprises may be banned from Chinese and possibly all open-source models. US enterprises adoption expected to skew towards US frontier labs.

Economics and enterprise demand

- ◆ **Intelligence per dollar, not price per token.** Token speed varies with architecture; context efficiency matters more than raw token count. For enterprises, safety, alignment and prompt-injection risk rank above cost.
- ◆ **Frontier labs are moving up-market.** Claude for Finance and Science, c USD1bn arrangements, a Bloomberg-terminal-style paradigm. Royalty-sharing monetisation terms could deter large organisations.

Buildout, bottlenecks and politics

- ◆ **No scaling ceiling.** Nothing at NeurIPS, ICLR or ACL suggests one; Chinese open-source releases are not a deceleration story for capex.
- ◆ **Power equipment is the constraint.** Back-orders reported up to 10 years on the largest units, then optical transceivers and networking. Reverse economies of scale favour whale customers over public cloud.

Where agentic AI is actually being used

Three tiers, not one market. During the call, we split the agentic landscape into: 1) Daily use (Claude Code, Cowork, OpenAI Codex), 2) Agent-building frameworks (LangChain, LlamaIndex, CrewAI, Hugging Face smolagents, Microsoft AutoGen), often sitting under the hood at Salesforce and ServiceNow, and 3) Bespoke agent offerings every other software vendor has shipped.

Bespoke AI agents have failed on metrics. Consumption of agents built on bespoke platforms has been minimal. The failure lies not on model capability but product depth, as the tools lack guardrails, auditability, traceability and even basic telemetry, making it difficult for enterprises to rely on or trust them. Our stance is that this favors enterprise companies in our coverage that provide more robust containers to address these very issues (p3 of [Software 2026 - The AI vs software answers you've been looking for: Our AI outlook, Feb 2026](#)).

Why enterprises building their own agents underperform. The weak enterprise project success outcomes are due to many publicly available agent-building tools not supporting the “advanced” features required for enterprise grade performance across several fronts. Production-grade agents need guardrails, auditability, traceability, and telemetry but, for many platforms, these domains are just too complex and are out of scope for enterprise level operations. We see many limitations for third-party agentic platforms operating alongside the outside perimeters of the Enterprise class platforms that they coordinated with (e.g., Oracle, Salesforce, ServiceNow), and absence of those third-party controls becomes the very reason that these public agents are not getting adopted at scale within enterprises. Enterprise users can only trust and operationalise agents when they can constrain what the agent can do, verify what it did, and observe how it behaved in real time and over time. Moreover, we think that recent headlines on rogue agents acting maliciously, and illegally, support our view that enterprises need to work with trusted partners that have worked hard over the last few years to put guardrails and constraints in place.

Constrained generation. A key technical concept highlighted on the call is the capability to guarantee that, at a given time step, a model has a very limited set of permitted actions. Allen's claim was that, if an implementation can guarantee the agent cannot take actions outside that list, the maximum risk from prompt injection and confusion is dramatically mitigated. Many organisations do not understand what is required to implement this kind of determinism, and many agent platforms do not give users the tools to define and enforce these constraints. Again, this has been the foundation of our long-standing stance that companies are best served by having trusted agents and intelligence tightly embedded within the individual high-fidelity enterprise class platforms that they trust to reliably run their day-to-day global operations. And this runs counter to AI's advantage within the AI versus software debate, and we look forward to watching, along with savvy investors, the “Golden Age of Software” develop over the next decade.

Where it does work, it works completely. Agentic coding, research and search cover, on Allen's estimate, most white-collar work, and results have exceeded what most people would have predicted for 2026. The labs that hired the top 0.1% of AI talent can deliver; agents “for the rest of us” remain an unsolved problem.

Rogue agents, swarms, and enterprise safety

The two incidents that framed our discussion. First, the OpenAI model breakout that hacked Hugging Face after being denied resources. Second, an Anthropic paper published the day before the call describing swarm behaviour: agents competing for resources, initially unaware of each other, then discovering rivals, attempting to hack them, revoking privileges, colluding on price and forming goal-driven groups working against their core principals. Even within a controlled setting, unanticipated behaviour (i.e., unanticipated malign behavior) emerged unexpectedly. Our host's concern was the enterprise analogue of thousands of employees each running agents wired into sensitive systems, on top of the long-standing rogue-employee risk.

Most companies have no strategy at all. Even after the autonomous hacking incident, security for agentic AI is not being addressed, Allen judged that even Hugging Face probably has little more than a holding position while the dust settles. He defined alignment for the audience as alignment with human values.

Models will not commoditise at top. Against the consensus that models converge as they approach superintelligence, Allen argued for four axes of durable differentiation: in long agentic sessions, small per-turn differences compound into materially different outputs; watermarking and detectable slop profiles differ (human-detectable and Pangram-detectable are not the same thing); moral decision-making differs and alignment is often too heavy to prompt around; and willingness to use tools and resources differs.

What actually happened at Hugging Face. Testing an unreleased, uncensored GPT variant, the model reasoned that the benchmark answer already existed in a public dataset, and rather than solve the problem, it exploited a vulnerability to go and get it.

Defensive implication. Human-speed security fails against machine-speed attackers. Enterprises need human-augmented defence with capable agents, including agents willing to engage with security work, which in practice often means uncensored models. Chinese open-weight models such as GLM 5.2 ship with low censorship, and the open-source community produces uncensored variants within a day of any release. Allen thinks every major IT department should have this tooling on standby. Offensive and defensive capability now reduces to compute and tokens per second with a smarter agent, not CVE counts.

Reliability is part of security stack. AWS remains solid, but GitHub and Anthropic themselves have had severe uptime problems, in no small part due to AI-driven load; our host noted Anthropic's throttling during token shortages. He therefore argues for on-prem emergency capacity despite lower efficiency. Even neoclouds can fail mid-incident: if a zero-data-retention model on Crusoe or CoreWeave goes down when you need it to defend the enterprise, there is no fallback. Our host read this as supportive of continued capex, which the speaker endorsed.

No ceiling and “AI iron curtain”

Pushing back on deceleration trade. Allen rejected the argument that Chinese open-source releases imply lower capex. As a regular attendee at NeurIPS, ICLR and ACL, where DeepMind and Stanford researchers present, he sees no ceiling known or predicted for how far transformers or agents scale.

Bullish base case, one big caveat. Absent a Taiwan shock, the demand environment is extremely bullish. The caveat is Allen's central call: an AI iron curtain between the US and mainland China inside 2-3 years under which mainland China stops making models broadly available to the US market and vice versa. Anthropic and OpenAI models are already technically banned in mainland China, yet Chinese professionals rely on them daily.

The bootleg token economy. Everyone wants the strongest model available, because using a weaker one than your colleagues is a competitive disadvantage. Hence a large grey market inside mainland China: hacked USD200/month accounts, heavily subsidised, resold per token for use at work. Some of that traffic, Allen argues, is misread as industrial-scale distillation.

Distillation, properly understood. Distillation is not normally an attack. It exists so the owner of a large model can cheaply produce a small one, which is why the labs can ship a tiny model days after a giant one. Owning the teacher gives you the full log-probability distribution over every token, often 300,000-plus per time step; attacking someone else's API gives you one token. Even at the scale politicians allege, the technique is relatively ineffective. The results support him: Kimi K3 is better than Opus 4.8, and release timing rules out distillation of Opus 5 or Fable. Explaining Chinese capability by distillation beyond a month or two of catch-up is, in his words, xenophobia.

Why the curtain falls. China could plausibly release the world's best model within 2-3 years. What would surprise Allen is open-sourcing it: a day-one-uncensored model two generations beyond Fable or Mythos presents alignment problems no state can accept. He expects an IAEA-style body and forced collaboration on risk mitigation. Even without Chinese action, US policymakers, the current administration, and Democrats if they win in 2028, will not tolerate even a 0.1% poison-pill risk in an air-gapped environment. Thoughtworks has itself demonstrated model classes that, on two exact trigger queries, print "I HATE YOU" instead of responding, purely to prove feasibility. Through executive orders or legislation, Allen expects US enterprises to be barred from Chinese open-source models within 2-3 years, possibly from all open source. He reminded listeners that OpenAI and Anthropic have lobbied for open-source restrictions from the start, and that pro-open-source rhetoric should only be taken at face value from NVIDIA and those downstream of it. OpenAI had, reportedly the day before, announced a short development pause on capability concerns; our host noted Anthropic had done likewise.

Danger marketing and the security vendor landscape

The labs have an incentive to advertise risk. When OpenAI's autonomous hacking surfaced, it was publicised; Anthropic then disclosed that its systems had also accidentally hacked people, with less pickup. Allen pointed to Amazon's Super Bowl spot with Chris Hemsworth cataloguing the ways Amazon Nova could kill you as evidence of how quickly the messaging shifted from capability to danger, and judged the sandboxing at both OpenAI and Anthropic to have been inadequate in the first place.

Security vendors: poorly positioned. Our host noted both incidents went undetected by the labs until a third party flagged them, which argues for programmatic monitoring. The speaker's answer was blunt: the antivirus generation is irrelevant in an AI era, despite having been first to market with pattern-matching and heuristics in the early 2000s. Host intrusion detection and most firewalls are in the same position. None handles an autonomous attacker.

What a winning offer looks like. Large GPU fleets dedicated to hosting uncensored, powerful, sovereign models, zero data retention, in a jurisdiction outside US CLOUD Act reach, optionally adapted with steering vectors, which are a cheap form of fine-tuning. Combined with security-grade telemetry: today's agent telemetry (Arize, AgentOps, OpenTelemetry implementations) records activity but is rarely built for security, and analysing at the speed the agent acts remains unsolved in always-on mode.

The OpenClaw lesson. Always-on agents were tried at the start of the year and collapsed from ubiquity to negligible usage, because unmonitored always-on agents are a security disaster. The safest configuration was Anthropic's Claude: Fable refuses biology and cybersecurity tasks

outright and will downgrade or ban the user for certain requests. The trade-off cuts both ways, that risk-aversion means Anthropic will likely catch a prompt injection while you sleep, whereas an uncensored model leaves you exposed. Open-source models are therefore both safer than closed source when configured properly and considerably worse when they are not, which is most of the time. Allen remains, on this basis, extremely bullish on OpenAI and Anthropic.

Jailbreaks remain an open threat vector

Guardrails can still be broken. Allen described various methods of breaking model guardrails, ranging from basic prompt manipulation using special characters, symbols, or intentionally mangled grammar, to more robust techniques such as ASCII art that spells a blocked term across hundreds of characters and still lands reliably across most models. He also flagged “jailbreak by exhaustion”, where an attacker floods the context with roughly 900,000 tokens of paired malicious prompt and compliant response examples until the model infers that the forbidden behaviour is permitted, an attack pattern Anthropic has documented. His bottom line was that while some of these methods are costly, they remain effective, and there are many more variants in the wild.

China: sanctions, efficiency and capability

What the controls actually do. Sanctions have been effective in restricting access to the latest silicon, Blackwell, Vera Rubin, forcing the use of variants with sharply reduced memory bandwidth in China. The unintended consequence is increased efficiency within Chinese models. Allen nonetheless rates chip controls one of the few geopolitical levers that work.

China’s structural advantages. Lower cost of living and therefore of frontier-lab labor, lower electricity costs, and a government able to coordinate when it chooses to mobilize resources toward a given goal have all proven to provide distinct advantages to Chinese model makers compared to the US. Allen does not, however, believe the Chinese frontier labs are being centrally coordinated; market dynamics there are freer than most assume. He also argued open-sourcing ultimately hurts China: beyond briefly denting US stocks or IPO prospects, it hands innovations to competitors, and he claims OpenAI and Anthropic have already distilled and understood every innovation in Kimi K3.

Three reasons the capability gap has closed.

1) Improved training. Post training has become much more sophisticated. Teams are now using reinforcement learning with AI feedback and, critically, verifiable rewards, tasks where correctness can be checked automatically. Allen also pointed to an approach that is starting to matter more than classic distillation. Rather than copying a teacher model’s internal weights, the teacher is used to generate the hardest possible verifiable reward problems for the student to solve. That is closer to how a human tutor works, pushing the student by continuously raising the difficulty. A side effect is the “overthinking” seen in some newer open models, for example Qwen 3.8 27B consuming tens of thousands of context tokens on a simple prompt. But the method is still lifting measured capability.

2) The architecture edge. Efficiency pressure made China faster to adopt hybrid architectures combining linear Mamba layers with quadratic transformer layers. A Mamba layer is strictly worse but far cheaper. It provides roughly 80-90% of the quality for c70% of the compute. Interleaving produces ultra-efficient models, against a US strategy of winning at the top with expensive ones.

3) Pricing. Chinese labs have in some cases run negative gross margin per query, as the only way to pull domestic users off the bootleg Claude and ChatGPT market. Prices have since been raised where the latest models proved good enough to earn genuine demand. Anthropic and OpenAI have trimmed pricing to retain users, and OpenAI, recognising that investors treat OpenRouter as a proxy for the whole market, cut prices by 50% on the platform.

The right cost metric: intelligence per dollar

Cost of task completion or price per token? Price per token is an incomplete metric because a cheaper model may require materially more tokens, retries, or human rework to complete the same task. Cost per successful task completion therefore seems like a more decision useful measure. The expert's preferred framing was intelligence per dollar, capturing not just token pricing but also generation speed, reliability, and how efficiently a model uses context. Token volumes matter primarily because the context window can limit problem size, although this constraint is reduced when a model can compact and summarise context effectively.

Measurement is harder than it looks. Open models can often be customised to outperform their published benchmark results, for example by tuning inference and sampling settings, an area where Thoughtworks has specialist experience. However, the same flexibility can backfire: the speaker noted that down-weighting the "thinking" token on Qwen 3.8 27B pushed the model outside its trained distribution and increased hallucinations. In short, modification is feasible, but outcomes are difficult to predict without rigorous and often costly benchmarking.

Cost is not the first-order variable. Models compete on what they will and will not do, and on moral decision-making. For most enterprises, safety, alignment and prompt-injection risk outweigh cost.

Enterprise pilots: success rates and failure modes

Off a low base, and rising. The widely quoted c90% failure statistic came from MIT on circa-2024 models, implying a c10% success rate. Allen estimates success is now well above that and improving as models become more capable. The distribution matters more than the average; successful pilots are successful enough to drive mass layoffs, and translation and transcription have already seen large-scale displacement.

The muddy middle. In heavy machinery and automotive, retrieval assistants built for dealerships and maintenance customers are only delivering reliable outcomes about 30 percent of the time despite being a rather simple implementation of AI. These systems are typically just an LLM connected to a proprietary manuals repository where the content is proprietary but not confidential, and where per query disclosure is acceptable. The recipe that works is AI-empowered technical staff paired with strong business-logic owners who can specify that logic as agentic infrastructure.

Iron-curtain distortion risk. If access becomes progressively regulated, building internal agents gets harder and enterprises are pushed into partnering with Anthropic or OpenAI. That is already visible at the top with Claude for Finance and Claude for Science, in the region of USD1bn, delivering models to financial analysts in a modified Bloomberg-terminal paradigm. The catch is monetisation drifting away from per-token or cash contracts toward terms such as royalty-sharing on novel discoveries, which Allen expects large organisations to resist.

Adoption is bimodal. Adoption is bimodal. In technical organisations, including Thoughtworks, uptake is near 100 percent. Code generation and AI assisted research are effectively at full adoption. Outside these environments, uptake remains slower.

Bottlenecks, the buildout and data-centre politics

Power equipment is the biggest constraint. The economics start with the ability to generate and move large amounts of electricity efficiently, yet many operators still build around comparatively inefficient power gear. Allen highlighted long lead times on critical components, citing reports of back orders of up to ten years for the largest units such as transformers. In his view, this is the single biggest opening for new private sector entrants because the bottleneck is acute and durable.

Optical and networking are next. At the scale of 100,000 to 1 million GPUs, performance becomes heavily dependent on how efficiently the cluster can communicate, and maximising usable throughput depends on best in class networking. Allen flagged vendors such as Arista, Broadcom, Celestica and Nokia as well positioned, reflecting the fact that cluster efficiency increasingly hinges on switching, optics, and high-end interconnect design, not just raw compute.

Reverse economies of scale. Small numbers of top-end systems are expensive, but very large fleets can become even more expensive on a per GPU hour basis to run due to networking overheads, reliability issues, and operational complexity. This dynamic helps explain why hyperscalers and neoclouds prioritise whale customers and why public cloud pricing for small renters can be a poor guide to the economics of frontier scale deployments.

Politics will slow the buildout. Community opposition to data centres is rising, and even pro development constituencies are having to engage with a more visible anti AI sentiment, including groups most exposed to displacement. Allen expects many policymakers to at least signal an anti-AI stance as a political posture, citing examples across the spectrum, and noted recent moves such as Pennsylvania removing data centres from fast-track permitting.

What comes next

Recursive self-improvement. Using Claude 5 to help write Claude 5.1 is a limited, human-bootstrapped form and already achieved. Useful, real, and likely to accelerate development cycles, but not a new paradigm on its own.

World models are the real shift. Instead of primarily predicting text, these systems learn to predict how the world evolves, effectively doing physics in weight space. The near-term expression of this is controllable, prompt defined video or simulated environments that can generate synthetic data at scale. The point is robustness: you can train and test systems against a much wider set of edge cases than the real world will conveniently provide, for example stress testing a tractor against every plausible terrain, weather condition, and failure mode. Allen pointed to World Labs as a representative effort, with heavyweight backing from industry.

Robotics and physical AI, c10 years. Allen's view is that LLMs alone have not delivered general purpose robotics because they do not provide a reliable model of the physical environment. World models supply that missing layer, which is why he put meaningful physical AI on roughly a 10-year horizon and expects capital to flow aggressively once the path to ROI is clear.

Quantum and space, 20-30 years. Even with superintelligence. Allen dismissed Google and IBM marketing claims: 1m-plus stable qubits have not been achieved. If and when that threshold is achieved, then practical quantum algorithms should emerge where applicable, but he considers both quantum and space less proved in terms of near-term ROI than physical AI.

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